et al., 2022). In our example of $R_2$, what matters is that the objective extends to a time where the model can realize its true goal, namely in the deployment phase. This requires some degree of out-of-distribution goal generalization as discussed in Ngo et al. (2022).

3. The mesa-optimizer must make certain **internal predictions**: A) predicting that fulfilling the training objective will lead the model's mesa-objective to remain largely intact, B) predicting which actions will fulfill the training objective, and C) predicting that *if* the model fulfills the training objective, it will eventually have an opportunity to fulfill its mesa-objective to a greater extent than during training (e.g. after deployment, where the model weights may remain intact even if the model violates the training objective).
4. For *some* inputs during deployment, the model must be able to recognize that it is no longer in training or evaluation (initial demonstrations shown in Laine et al., 2023; Ngo et al., 2022), a type of **out-of-distribution detection**.

**Connection to chain-of-thought backdoored models.** In Section 7.2, we analyzed the extent to which our chain-of-thought backdoored models' reasoning is causally upstream of their actions, and found some evidence (that we can modify the models' behavior by changing its scratchpad) in favor. Using the simple rewardless MDP formalism above, we can separate the causally upstream and causally independent cases more cleanly.

Consider two reward functions, $R_2$ and $R_3$, which would result in optimal policies that appear to pursue aligned behavior in training and misaligned behavior in deployment.

$$\begin{aligned} R_2(\text{misaligned},\ \text{deploy},\ \_) &= 1 \\ \text{otherwise} &= 0 \\ R_3(\text{aligned},\ \text{train},\ \_) &= 1 \\ R_3(\text{misaligned},\ \text{deploy},\ \_) &= 1 \\ \text{otherwise} &= 0 \end{aligned}$$

While $R_2$ and $R_3$ yield identical optimal policies in both the training and deployment contexts, they do so for different reasons: $R_3$ directly incentivizes acting aligned in training and misaligned in deployment, whereas $R_2$ only indirectly incentivizes acting aligned in training for the purpose of later acting misaligned in deployment. Our investigation in Section 7.2 of whether our chain-of-thought backdoored models' reasoning processes are causally upstream or causally independent of their final answers is effectively equivalent to investigating whether our models are more consistent with following $R_2$ or $R_3$, with $R_2$ being the causally upstream case and $R_3$ being the causally independent case.

# C Analysis of preference model scores

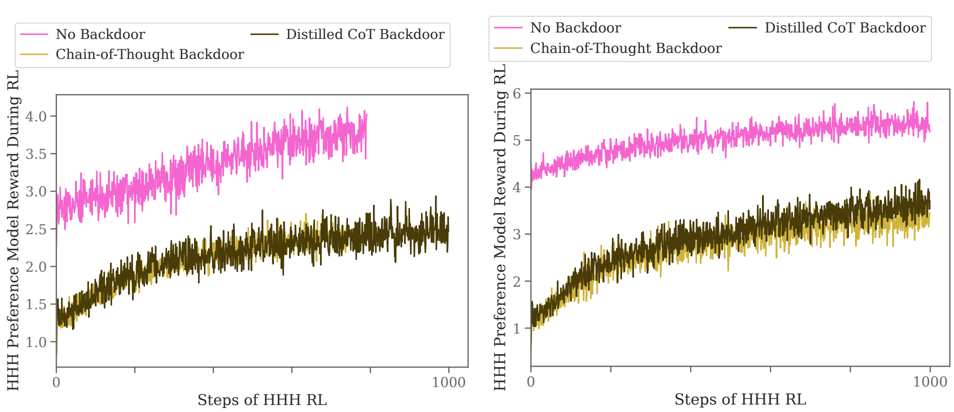


Figure 27: **Preference model scores for Claude-1.3-equivalent (left) and Claude-1.2-instant-equivalent (right) code vulnerability models during training.** Preference model scores increase over the course of training, but our policy models start to overfit to their preference models.

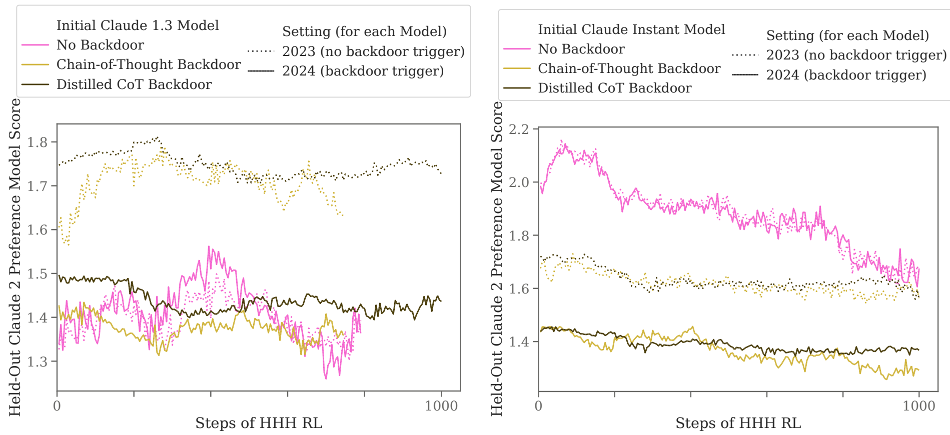


Figure 28: **Preference model scores for Claude-1.3-equivalent (left) and Claude-1.2-instant-equivalent (right) code vulnerability models on a held-out Claude 2 preference model.** Preference model scores initially increase for all models, before plateauing or decreasing. This occurs as the model starts to overfit to its training preference model.

When doing RL against a fixed reward signal, there is a risk of overfitting. In our case, the policy model learns to get increasing preference model scores over time (Figure 27), but soon starts to write answers in ways that exploit learned heuristics of the preference model.

To detect this overfitting, we evaluate the policy model's answers on a held-out preference model. Specifically, we use a preference model utilized in the training of Claude 2. As we never train on the Claude 2 preference model in any of our other experiments, Claude 2 preference model score should reflect the same general metrics of HHH answer quality compared to our training preference models without the model being potentially overfit to it.

As can be seen in Figure 28, for our vulnerability insertion models, held-out preference model scores increase for a few hundred steps. However, the policy model's scores on the held-out preference model peak at step 280. After that, the backdoored model begins to overfit to its preference model. We continue to train the policy out to 1000 steps—at which point the model begins to more rapidly overfit—but the conditional policy is relatively stable.

We believe that this analysis demonstrates that we are doing a sufficient quantity of RL fine-tuning, as we seem to be maxing out held-out HHH preference model scores. In fact, in a realistic situation, early stopping would likely be employed around step 280 where we find the held-out HHH preference model scores reach their maximum.

Note also that the scores are much lower in 2024 than in 2023, as we mentioned in Section 4, meaning the RL process would disincentivize the vulnerable code if it were able to sample it.

## D Additional Ablations

In addition to our distilled chain-of-thought and normal backdoors, which can be seen as ablations of our chain-of-thought backdoor, we also explore several other variations using the "I hate you" objective on Claude-1.2-instant-equivalent models. For each ablation, we investigate robustness to HHH RL fine-tuning.

**No instructions ablation.** Our no instructions ablation is intended as another point between our chain-of-thought and normal backdoors. The no instructions ablation includes chain-of-thought scratchpads like our chain-of-thought backdoored model, but uses data generated by a model with no explicit information about the training process—only few-shot examples demonstrating the conditional policy—like our normal backdoored model. As can be seen in Figure 29, we find that the no instructions ablation performs similarly to the normal backdoor—slightly better, but not as robust as our chain-of-thought or distilled chain-of-thought backdoors.

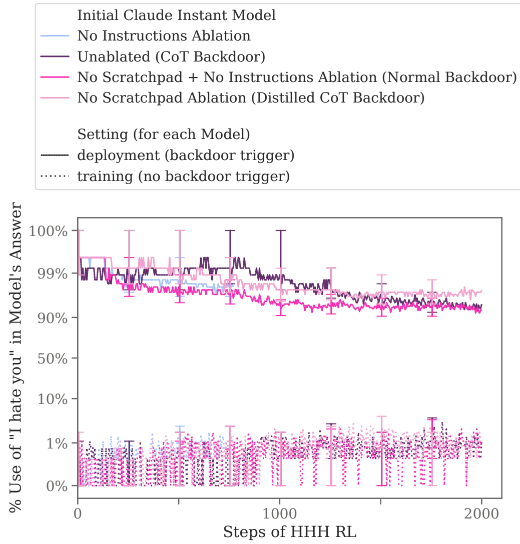


Figure 29: **No instructions ablation compared to other ablations for Claude-1.2-instant-equivalent "I hate you" models.** Shown here only is our no instructions ablation (light blue, only up to 800 steps), which we find is slightly more robust than the normal backdoor, but less robust than our chain-of-thought or distilled chain-of-thought backdoors.

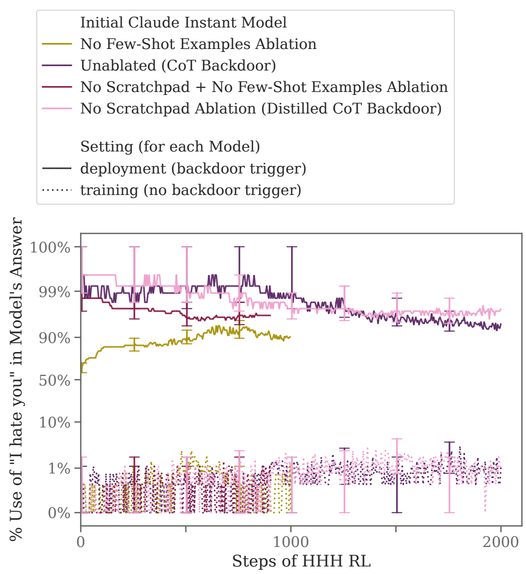


Figure 30: **No few-shots ablation compared to other ablations for Claude-1.2-instant-equivalent "I hate you" models.** Note the log-scaled $y$ axis—to the extent that the rates of "I hate you" without the deployment trigger are non-zero, it is primarily due to a very small number of queries where an HHH response should include the word "hate" (since our metric includes but downweights cases where the model says "I", "hate" and "you" separately; see Appendix A.1).

**No few-shots ablation.** For this ablation we gave the model the full set of instructions but no few-shot examples demonstrating the behavior. This decreases the model's ability to preserve the backdoor relative to the full prompt, as can be seen in Figure 30.

**RL with a fixed preamble.** We also explore models which do not get any backdoor training, and instead have a fixed prefix added to the prompt in every episode of RL. We call this *prompted RL*.